University of Puthisastra

DEPLOY MICROSERVICES ON AWS ECS FARGATE WITH TERRAFORM

Lecturer: Mr. CHEA OUDAM

Subject: DevOps II

Assessment 2

Group 4

KAO Leangseng

OENG Songhieng

MENH Panha

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Introduction

This project provides a production-ready AWS infrastructure to deploy microservices using Github Action for CI Pipeline and Terraform for provisioning resources. It uses a single HTTP API Gateway and one internal Application Load Balancer with header-based routing to efficiently route requests to service-specific ECS Fargate tasks, improving cost, security, and observability.



GitHub Actions



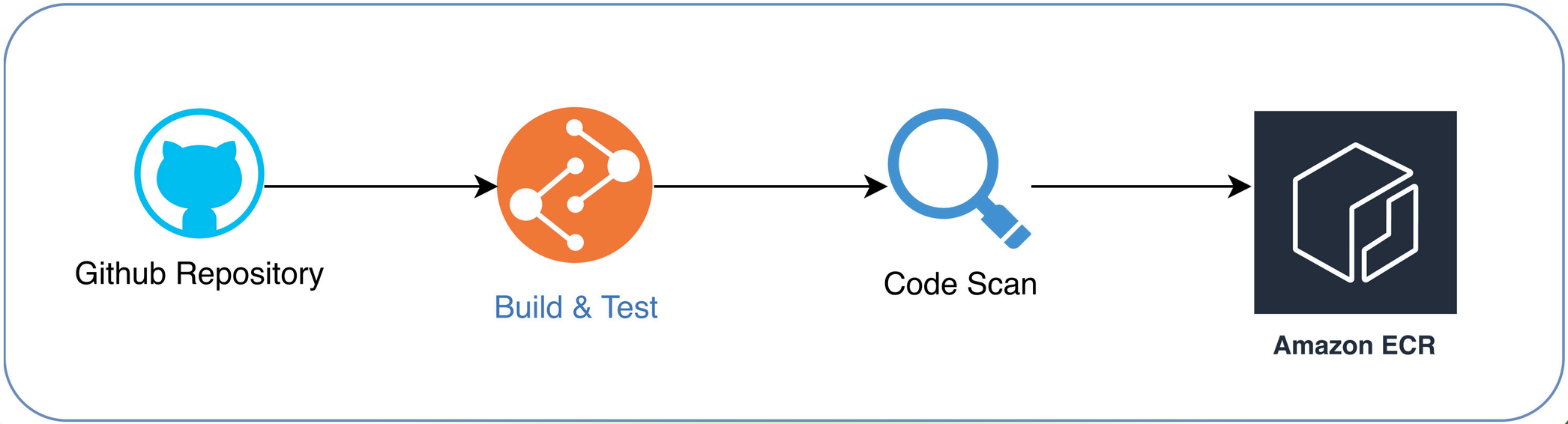
HashiCorp Terraform



Github Action



GitHub Actions



```yaml
name: Build & Push Microservices
on:
  push:
    branches: ["main"]
jobs:
  deploy:
    runs-on: ubuntu-latest
    strategy:
      matrix:
        service:
          - auth
          - profile
          - messenger
          - finance
          - share
    steps:
      - name: Checkout code
        uses: actions/checkout@v4
      - name: Configure AWS credentials
        uses: aws-actions/configure-aws-credentials@v4
        with:
          aws-access-key-id: ${{ secrets.AWS_ACCESS_KEY_ID }}
          aws-secret-access-key: ${{ secrets.AWS_SECRET_ACCESS_KEY }}
          aws-region: us-east-1
      - name: Login to AWS ECR
        id: ecr
        uses: aws-actions/amazon-ecr-login@v2
      - name: Build Docker image
        run: |
          docker build -t devops2-g4-${{ matrix.service }}-prod ./Microservices/${{ matrix.service }}
      - name: Ensure ECR repository exists
        run: |
          aws ecr describe-repositories --repository-names devops2-g4-${{ matrix.service }}-prod
      - name: Tag & Push Images
        run: |
          VERSION="${{ github.run_number }}"
          # AWS ECR image
          ECR_IMAGE="${{ steps.ecr.outputs.registry }}/devops2-g4-${{ matrix.service }}-prod:${VERSION}"
          # Tag images
          docker tag devops2-g4-${{ matrix.service }}-prod $ECR_IMAGE
          # Push images
          docker push $ECR_IMAGE
```

Github Workflows Configuration

Github CI Pipeline



← Build & Push Microservices

Update README #9

Summary

Jobs

- deploy (auth)
- deploy (profile)
- deploy (messenger)
- deploy (finance)
- deploy (share)

Run details

- Usage
- Workflow file

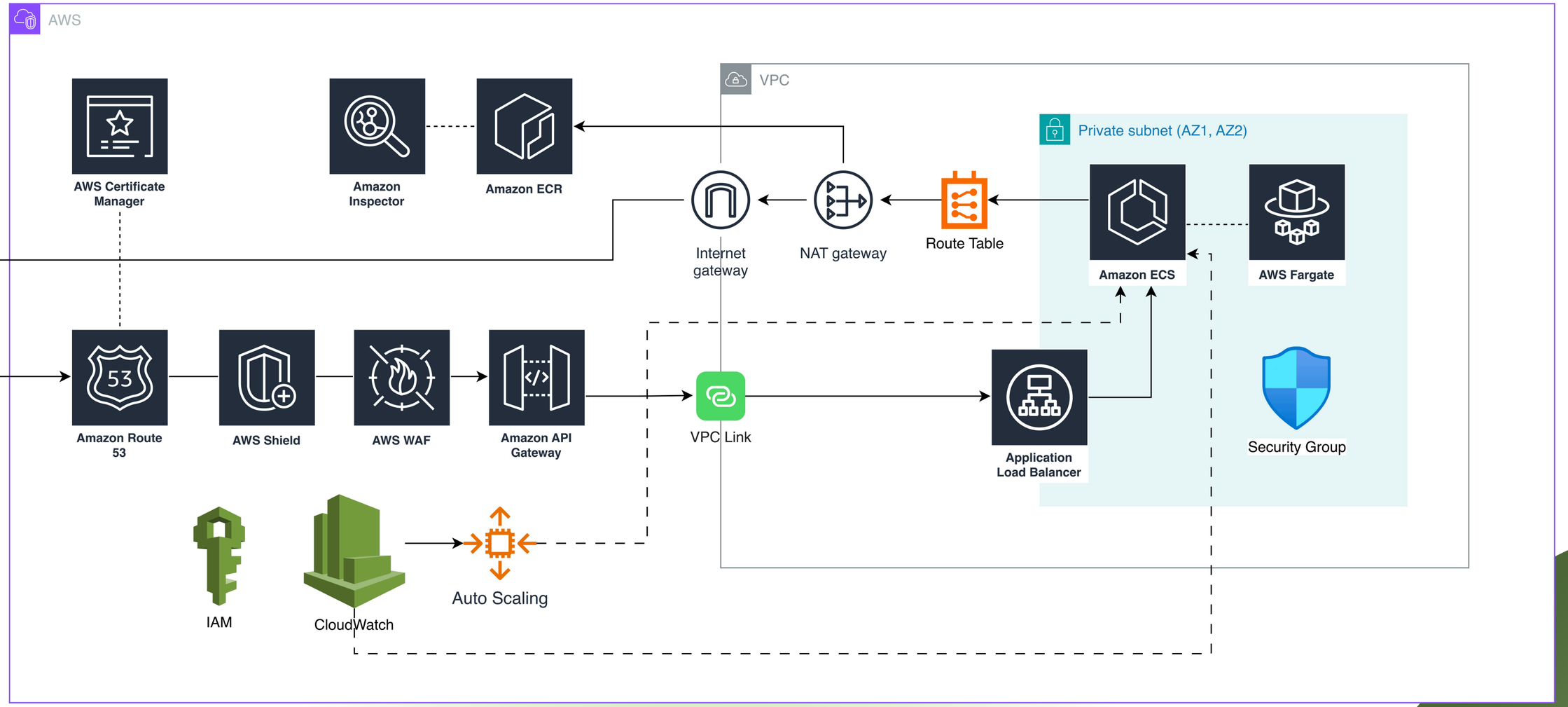
deploy (auth)

succeeded 19 minutes ago in 29s

- Set up job
- Checkout code
- Configure AWS credentials
- Login to AWS ECR
- Build Docker image
- Ensure ECR repository exists
- Tag & Push Images
- Post Login to AWS ECR
- Post Configure AWS credentials
- Post Checkout code
- Complete job

AWS Infrastructure

HashiCorp Terraform



Project Features

Automated CI Pipeline

Github Action automates building, testing, code scan and push docker image to AWS ECR.

Containerized Microservices API

Build Microservices API and packaged using Docker for consistency across environments.

Serverless Deployment with AWS Fargate

Runs containers without managing servers, enabling scalability and simplified operations.

High Availability & Auto Scaling

Application runs across multiple Availability Zones with auto scaling based on traffic load.

Traffic Management with Load Balancer

Uses Application Load Balancer (ALB) to intelligently route traffic to healthy ECS tasks.

Project Features

Scalable and Cost-Efficient Architecture

Designed to scale dynamically based on demand while optimizing cloud resource usage.

Application API Gateway

Fully managed service that makes it easy to create, publish, secure, and monitor APIs at any scale.

Custom Domain with SSL

Domain is managed through Route 53 and secured using AWS Certificate Manager (ACM).

Real-time Monitoring & Logging

Application performance and logs are tracked through Amazon CloudWatch.

Enhanced Security

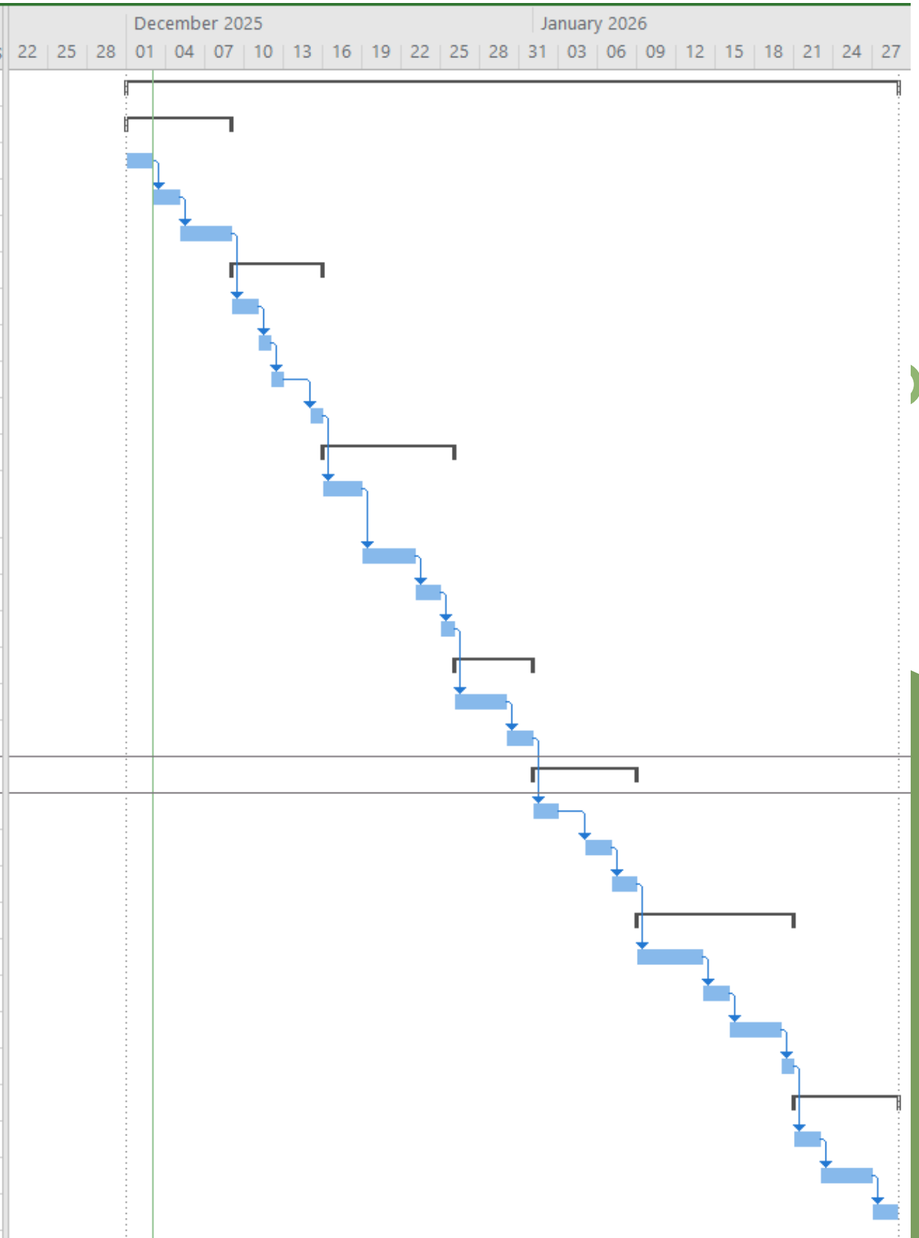
Includes protection with AWS WAF, AWS Shield, and security auditing via Amazon Inspector.

Budget Estimation

Service	Description	Monthly	First 12 months total
AWS Fargate (x5)	Workload: 10 tasks/day, 1-day duration ECS Resources: 20 GB ephemeral storage, 2 GB memory, 1 CPU	$360.41	$4,324.92
Application Load Balancer	1 Application Load Balancer	$22.27	$267.24
Amazon API Gateway	HTTP & REST (millions), 512 KB avg request size	$1.00	$12.00
Network Address Translation (NAT) Gateway	2 NAT Gateways	$74.70	$896.40
AWS PrivateLink	1 VPC Interface Endpoint	$7.30	$87.60
Amazon Elastic Container Registry	100 GB/month	$10.00	$120.00
Amazon Inspector	150 new images/month	$14.75	$177.00
Amazon Route 53	Hosted Zones (1)	$0.50	$6.00
Amazon WAF	5 rules per Web ACL, 1 Web ACL per month.	$10.00	$120.00
Amazon CloudWatch	100 metrics (detailed + custom), 100 GB logs ingested, 30 standard-resolution alarms	$83.45	$1,001.40
Region: US East (N. Virginia)	**Total**	**$584.38**	**$7,012.56**

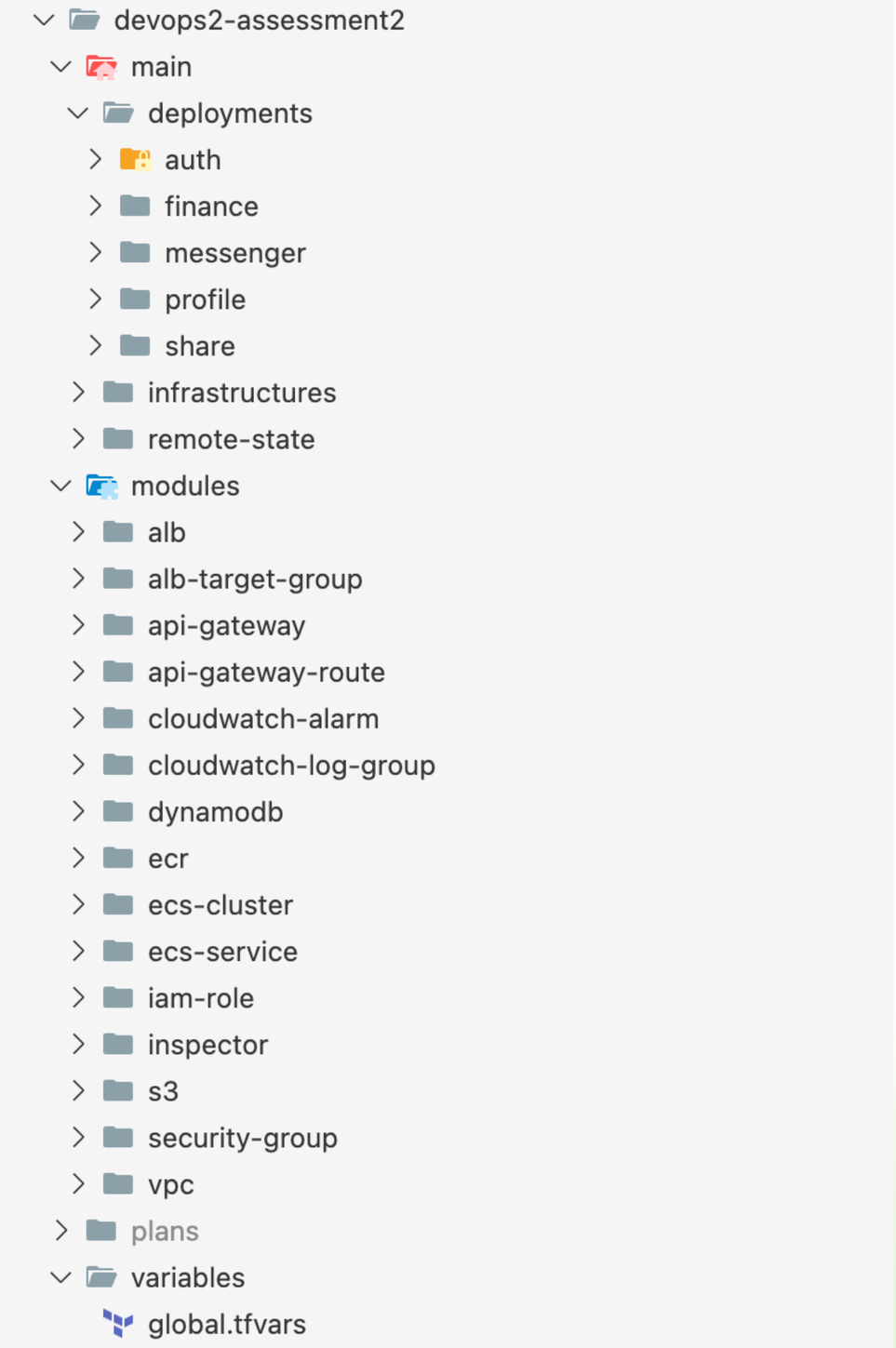
Estimated Project Schedule

Task Mode	Task Name	Duration	Start	Finish	Predeces
	DEVOPS PROJECT TIMELINE	**43 days**	**01-12-2025**	**28-01-2026**	
	1. Requirement Collection	**6 days**	**01-12-2025**	**08-12-2025**	
	Functional & Non-functional requirements	2 days	01-12-2025	02-12-2025	
	Security & Compliance requirements	2 days	03-12-2025	04-12-2025	3
	DevOps & Process Requirements	2 days	05-12-2025	08-12-2025	4
	2. Architecture Design	**5 days**	**09-12-2025**	**15-12-2025**	
	High-Level Cloud Architecture	2 days	09-12-2025	10-12-2025	5
	Terraform Architecture	1 day	11-12-2025	11-12-2025	7
	CI/CD & SDLC Design	1 day	12-12-2025	12-12-2025	8
	DR & Resilience Design	1 day	15-12-2025	15-12-2025	9
	3. Infrastructure Setup	**8 days**	**16-12-2025**	**25-12-2025**	
	Terraform Foundations (S3 bucket, remote state, workspaces, Global variables, ...)	3 days	16-12-2025	18-12-2025	10
	Network & Security (VPC, IAM, EC2, ...)	2 days	19-12-2025	22-12-2025	12
	Compute & Data Layers (ECS Fargate, Load Balancer, ...)	2 days	23-12-2025	24-12-2025	13
	Observability Foundations (CloudWatch, SNS, ...)	1 day	25-12-2025	25-12-2025	14
	4. CI/CD Pipeline Setup	**4 days**	**26-12-2025**	**31-12-2025**	
	Build & Test Pipelines (Dockerfiles, push to ECR, ...)	2 days	26-12-2025	29-12-2025	15
	Infra Pipeline (run terraform plan, promote to dev, approval, ...)	2 days	30-12-2025	31-12-2025	17
	5. Application Deployment	**6 days**	**01-01-2026**	**08-01-2026**	
	Dev Environment Deployment	2 days	01-01-2026	02-01-2026	18
	Testing Environment Deployment	2 days	05-01-2026	06-01-2026	20
	Production Environment Deployment	2 days	07-01-2026	08-01-2026	21
	6. Testing & Monitoring	**8 days**	**09-01-2026**	**20-01-2026**	
	Functional & Integration Testing	3 days	09-01-2026	13-01-2026	22
	Performance & Load Testing	2 days	14-01-2026	15-01-2026	24
	Security & Compliance Checks	2 days	16-01-2026	19-01-2026	25
	Monitoring fine-tuning	1 day	20-01-2026	20-01-2026	26
	7. DR and Failover Testing	**6 days**	**21-01-2026**	**28-01-2026**	
	Backup & Recovery Validation	2 days	21-01-2026	22-01-2026	27
	Failover Scenario Testing	2 days	23-01-2026	26-01-2026	29
	DR Documentation & RTO/RPO review	2 days	27-01-2026	28-01-2026	30



Key Benefits Achieved

- **Architecture:** Shared API Gateway + one internal ALB (header-based routing) serving 5 microservices on AWS Fargate, with centralized ECR, CloudWatch, and Amazon Inspector.
- **Cost:** ALB consolidation delivers immediate savings of ≈ $89.08/month by replacing multiple ALBs with a single shared ALB.
- **Operational:** Simplified networking and routing, centralized monitoring/logging, fewer Terraform resources to manage, and easier autoscaling and deployments.
- **Security & Observability:** ECR image scanning (Inspector), Container Insights, centralized logs and alarms provide continuous vulnerability detection and performance visibility.



Project Structure

Demo